Blueprint

The Inward Bow of a Real-Rooted Polynomial: Newton’s Inequalities, Machine-Checked in Lean 4

Inequalities for polynomials · real-rooted polynomials and log-concavity

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Statements

Theorem 1 — Newton’s inequalities, in Lean 4; Newton.newton_inequality 3
Lemma 1 — RealRooted.derivative , RealRooted.iterate_derivative 4
Lemma 2 — RealRooted.reverse 5
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At a glance

What the paper contributes

A machine-checked, `sorry`-free proof in Lean 4 over Mathlib of **Newton's inequalities** (1707): for a real-rooted real polynomial of degree n, $e_k^2 \cdot C(n,k-1) \cdot C(n,k+1) \geq e_{k-1} \cdot e_{k+1} \cdot C(n,k)^2$ for $1 \leq k \leq n-1$; equivalently the normalized means $p_k = e_k / C(n,k)$ are log-concave.

- **The theorem** (Theorem 1): the symmetric-function form assumes p monic; the coefficient form `newton_inequality_coeff` assumes nothing.
- **The route** is the classical one (Sylvester; Hardy–Littlewood–Pólya): differentiate, reverse, differentiate again, down to a real-rooted quadratic, whose discriminant is the inequality once positive factorials are cleared.
- **Four reusable blocks**: real-rootedness under differentiation, iterated differentiation and reversal, and the coefficient-form discriminant for degree ≤ 2, which also absorbs the degenerate case $a_{i-1} = 0$.
- **Boundary**: the mathematics is classical; the paper claims the formalization (to the author's knowledge the first in any proof assistant, June 2026). The equality cases and Maclaurin's chain are named, not proved.

🚀 **Start here: Theorem 1**: Newton's inequalities for a monic real-rooted polynomial, and log-concavity of the normalized means.

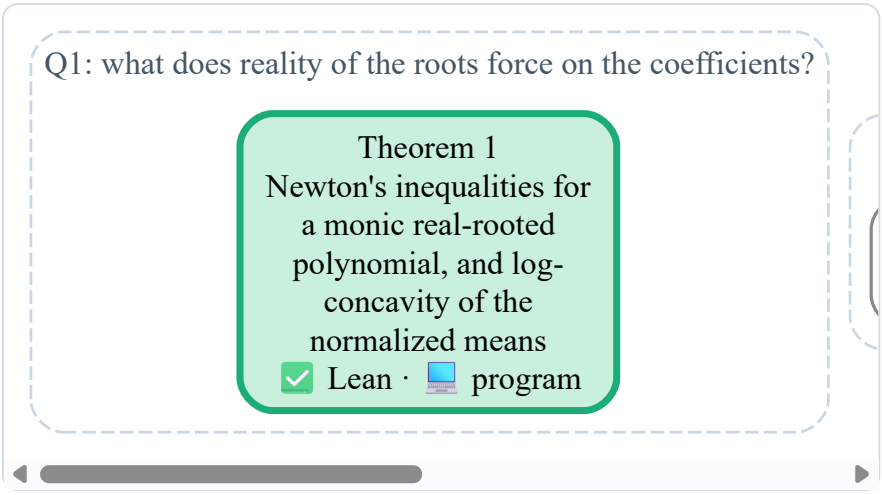
Main results:

- **Theorem 1**: Newton's inequalities for a monic real-rooted polynomial, and log-concavity of the normalized means

4 results | ✅ **1** formalized in Lean | ◐ **0** with a part in Lean | 💻 **1** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's “Evidence & scope” block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 **Setting (copied from the paper, §1)**

Write a real-rooted polynomial through its roots, $p(x)=\sum_{k=0}^{n}(-1)^k e_k\,x^{n-k}$, where e_k is the k-th *elementary symmetric function* of the roots, and form the *normalized means* $p_k=e_k/\binom{n}{k}$. *(§2)* I work over $\mathbb{R}$. A polynomial is *real-rooted* () when it splits into linear factors over $\mathbb{R}$—Mathlib's intrinsic ; equivalently it has $\deg p$ real roots counted with multiplicity. For the roots multiset $s=p.\texttt{roots}$, the elementary symmetric function is $e_k=\backslash\text{lean}Multiset.esymm\;s\;k$, and $\binom{n}{k}=\backslash\text{lean}Nat.choose$.

1 Q1: what does reality of the roots force on the coefficients?

Theorem 1. ✅ Lean **(Newton's inequalities, in Lean 4; `Newton.newton_inequality`)** ✓ p. 2

> *For a monic real-rooted real polynomial p of degree n and every $1\le k\le n-1$,*
>
> $$e_k^2\binom{n}{k-1}\binom{n}{k+1}\;\ge\;e_{k-1}\,e_{k+1}\binom{n}{k}^2.$$
>
> *Equivalently (`newton_logConcave`) the normalized means $p_k=e_k/\binom{n}{k}$ are log-concave, $p_{k-1}\,p_{k+1}\le p_k^2$. Both are `sorry`-free; `#print axioms` on each reports only `propext`, `Classical.choice`, `Quot.sound`.*

✅ Checked when this page was built: NewtonInequalities.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — newton_inequality, newton_logConcave, newton_inequality_coeff: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean NewtonInequalities.lean` (SHA-256 of the compiled file: `69dd05da620a4eb0…`; a source page on this site may show it with comments edited, under its own hash).

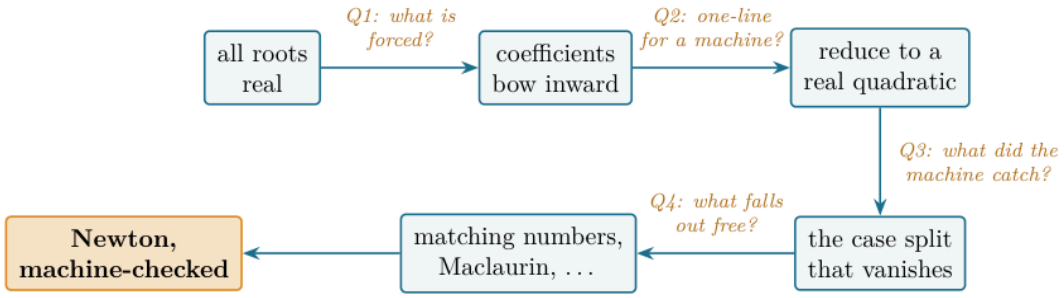
🆕 **Classical and new:** classical: stated by Newton (1707), proved by Sylvester, textbook account in Hardy–Littlewood–Pólya; the paper claims the formalization, to the author's knowledge the first in any proof assistant (searches of June 2026, §6).

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

Figure 1



The reader's map. From a hypothesis on the roots to a certified inequality and its corollaries; each arrow is the question its section answers. Every node is a block of `sorry`-free Lean.

Evidence & scope

- **Program:** the worked instances: e_k of (x−1)(x−2)(x−4), the Newton gaps 21 and 84, the zero gaps of $(x-1)^4$ and the gap −4 of x^2+1 (numpy, with tolerance); script in the paper repository, not in the Zenodo record. `figures/newton_fig1_bow.py`

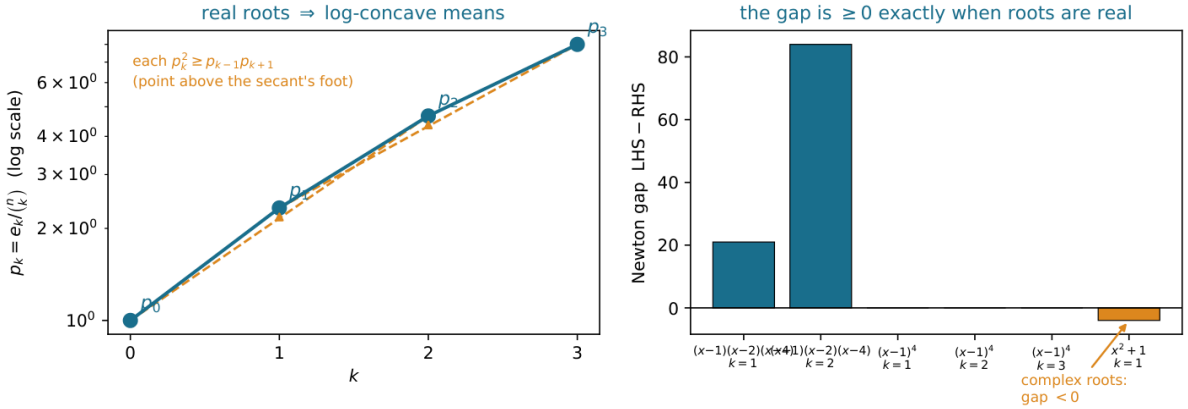
5 The building blocks

Lemma 1. (`RealRooted.derivative`, `RealRooted.iterate_derivative`) p. 6

> *If p is real-rooted then so is p', and hence every iterated derivative $p^{(k)}$.*

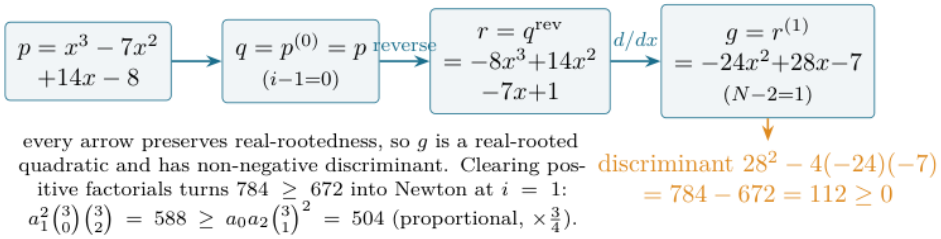
↳ Not used by another result of the paper.

Figure 2



Left: the normalized means of $(x-1)(x-2)(x-4)$, plotted on a log axis; each interior mean sits above the geometric mean of its neighbours (the amber arrows), so the sequence is concave—it "bows inward." *Right:* the Newton gap $e_k^2\binom{n}{k-1}\binom{n}{k+1} - e_{k-1}e_{k+1}\binom{n}{k}^2$ at every interior k: positive for the real-rooted cubic, exactly zero for $(x-1)^4$ (the sharp case), and *negative* for x^2+1. The generating script asserts every plotted value against exact integer arithmetic.

Figure 3



The differentiate–reverse reduction, worked on the running cubic at $i = 1$ (relating $a_0, a_1, a_2 = -8, 14, -7$). The final object is a quadratic; a real-rooted quadratic has non-negative discriminant; the factorials the proof cancels are the proportionality constant.

Lemma 2. (`RealRooted.reverse`) p. 6

> *If p is real-rooted then so is its reversal p^{rev}.*

↳ Not used by another result of the paper.

Lemma 3. (`realRooted_discrim_coeff`) p. 6

> *A real-rooted q with* $\deg q \leq 2$ *satisfies* $q.\mathtt{coeff}\ 1^2 \geq 4\, q.\mathtt{coeff}\ 2 \cdot q.\mathtt{coeff}\ 0$.

↳ Not used by another result of the paper.

References

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I. Newton, *Arithmetica Universalis* (1707); the inequalities among the coefficients of a polynomial with real roots are stated there without proof. 🔎 find it

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C. Maclaurin, A second letter … concerning the roots of equations, with the demonstration of other rules in algebra, *Phil. Trans. Roy. Soc. London* **36** (1729) 59–96. 🔎 find it

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R. P. Stanley, Log-concave and unimodal sequences in algebra, combinatorics, and geometry, *Ann. New York Acad. Sci.* **576** (1989) 500–535. 🔎 find it

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